Practice workbook · Utility software: anti-malware, defrag and file tools

Purpose and functions of utilities: anti-malware types and features, defragmentation, backup, repair and conversion, including choosing a utility for an issue and showing a disc after defragmenting.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-13

1. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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1. Question 5 · part c · 4 marks

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Worked-example retry · Attempt part c only.

(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

Issue	Utility software
Her word processor cannot read all of a file because it is corrupt.	
She has a JPEG image and needs to send it to a publisher who can only read PNG image files.	
There has been a flood and files on a server have been damaged, but Anna needs to access her data.	
Her hard drive is running very slowly when she tries to access a file.	

(4)

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1. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

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Justification

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(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

(Total for Question 5 = 15 marks)

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1. Question 5 · part c · 4 marks · Working and answer

Working and answer

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2. Question 3 · part e · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part e only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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2. Question 3 · part e · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part e only.

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(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

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Working and answer

3. Question 6 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

2 .....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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P 6 9 2 9 7 A 0 1 8 2 0

3. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 6 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

2 .....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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4. Question 6 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

(4)

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4. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part d · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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5. Question 3 · part d · 2 marks

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Further practice · Attempt part d only.

(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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P 7 2 9 3 7 A 0 9 2 4

5. Question 3 · part d · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

(2)

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(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

(4)

Task	Translator
A guessing game that can be used on different computing platforms	
A screen driver for a new smartphone	
A new version of a spreadsheet program for sale next year	
Control software for an embedded system inside a new washing machine	

(Total for Question 3 = 15 marks)

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5. Question 3 · part d · 2 marks · Working and answer

Working and answer

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